Midterm 2017, "The Mathematical Foundations of Political Methodology"

You have 90 minutes for this exam. You may use a calculator and refer to your Pemberton & Rau book. Please do not use the internet during the exam. Email your solutions, *with your work shown*, to Milena. Solutions without work will get no credit.

1. Given the following hold, what do we know about the relationship between sets A and C?

   (a) $A \cap B = B$ and $B \cap C \neq \emptyset$

   ANSWER: $A \cap C \neq \emptyset$.

   (b) $A \cup B = B$ and $B \cap C \neq \emptyset$

   ANSWER: Nothing.

   (c) $B \cap D = C$ and $B \cap C = A$

   ANSWER: $A = C$.

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a)
   $$\frac{4n^2 - 1}{(n+1)^2}$$

   ANSWER: It's 4.

   (b)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{7 + n^{\frac{3}{2}}}$$

   ANSWER: It's 0.

   (c)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{4}} + n^{\frac{1}{2}}}$$

   ANSWER: It doesn't exist.

3. Use the (δ, ϵ) definition of a limit to prove that the function

$$f(x) = \frac{2}{3}x - 17$$

is everywhere continuous.

ANSWER: We need that the limit as $x \to x_o$ of $f(x)$ equals $\frac{2}{3}x_o - 17$ for all x_o. For any $\epsilon > 0$, we can find a $\delta > 0$ such that $|\frac{2}{3}x - 17 - \frac{2}{3}x_o + 17| < \epsilon$ whenever $|x - x_o| < \delta$. Rearranging, we get $\frac{2}{3}|x - x_o| < \epsilon$ when $|x - x_o| < \delta$. Setting $\delta = \frac{3}{2}\epsilon$ proves the result.

4. Let

$$f(x) = \frac{x}{x^2+1} + 5$$

(a) Find the critical points of f and characterize them as maxima, minima, or inflection points.

ANSWER: Critical points at ± 1; 1 is a local maximum, -1 is a local minimum.

(b) Find the regions on which f is convex and concave.

ANSWER: f is concave on $(-\infty, -\sqrt{3})$, convex on $(-\sqrt{3}, 0)$, concave on $(0, \sqrt{3})$, and convex on $(\sqrt{3}, \infty)$. We can find this by evaluating the second derivative.

(c) Find any asymptotes that f may have.

ANSWER: As $x \to \infty$ f approaches the line $y = 5$ (from above). As $x \to -\infty$ f approaches the line $y = 5$ (from below).

5. Perform the following matrix operations:

(a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

ANSWER: It is

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find the rank of

$$\begin{bmatrix} 1 & 2 & 5 \\ -3 & 2 & -1 \\ 0 & 4 & 7 \\ 2 & 0 & 3 \end{bmatrix}$$

ANSWER: It is 2.

(c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.

$$\begin{bmatrix} 1 & 3 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

ANSWER: It is none of the above (the 2X2 principal minors are different signs).

6. Find and classify all of the critical points of the following functions:

(a)

$$f(x,y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

ANSWER: There are a continuum of critical points: all points of the form $(C, 5C)$ with $C \in \mathbb{R}$. They are all global optima, because f is globally concave: Its Hessian is

$$\begin{bmatrix} -20 & 4 \\ 4 & -\frac{4}{5} \end{bmatrix}$$

so it has determinant 0 and its diagonals are negative. Thus the Hessian is negative semidefinite.

(b)

$$f(x,y,z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

ANSWER: There is one critical point: $(\frac{1}{\sqrt{2}}, 1, \frac{1}{2})$. It is a global optimum because the function is the sum of concave functions of one variable. No Hessian required.

(c)

$$f(x,y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$

ANSWER: The Lagrangian is

$$L(x, y, \lambda) = x^2 + y^2 - xy - \lambda(2x - y - 3).$$

The first order conditions imply that $2x - y - 2\lambda = 0$, $2y - x + \lambda = 0$, and $2x - y - 3 = 0$. Solving for x and y we get $(\frac{3}{2}, 0)$ as a critical point (incidentally, $\lambda = \frac{3}{2}$). It is a constrained (global) minimum. We know this because $x^2 + y^2 - xy$ is globally convex and our constraint is linear. The Hessian of f is

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

which has determinant $3 > 0$ and positive diagonals.